Chapter 10. Mass-storage structure. Exercises

Original problems. State assumptions, units, and intermediate reasoning. Answers are in the separate solutions PDF.

OS10-E01. Disk scheduling

The head starts at cylinder 50, initially toward larger numbers. Pending requests are 82,170,43,140,24,16,190.

1. Give FCFS order.
2. Give SSTF order starting from 50.
3. Explain SSTF starvation.

Check your approach. Show the service sequence and the distance calculation under the stated model.

OS10-E02. RAID 5 and mirror groups

Compare four equal disks used as one RAID 5 array with the same disks used as two independent two-disk RAID 1 mirrors. Count both mirror groups' total capacity. No striping between mirror groups is assumed; ignore controller cache and scrubbing.

1. What is the RAID 5 data capacity?
2. What is the total mirror-group data capacity?
3. Compare one-disk fault recovery and state the conditions.

Check your approach. Keep array organization, capacity, and tolerated failure patterns distinct.

OS10-E03. A stable write

One second after a write reports success, power fails. On restart the final 512 bytes are missing.

1. What did success promise if the call only updated the page cache?
2. What can establish a stronger durability condition?
3. Is a mirror sufficient by itself?

Check your approach. Distinguish cache acceptance, device acknowledgment, and durable commit.

OS10-E04 RPM, IOPS, and response

Teaching HDD: seek 5 ms, 7200 RPM, uniform target phase, 64 KiB requests, 128 MiB/s transfer, and no cache. The specified request waits 15 ms before service. For the separate service-limited IOPS estimate, assume zero queue wait, a continuously supplied workload, and serial service without overlap.

1. Find rotation, transfer, and service.
2. Find the response time for the specified request; separately find service-limited IOPS under the zero-queue serial model.
3. Find bandwidth at 1000 IOPS for 4 KiB and 64 KiB.

OS10-E05 Flash mappings and WA

4 KiB pages, four-page erase blocks. B0:P0…P3 contain L0…L3; B1:P4…P7 are erased. Program new L1 into P4 and update its mapping before reclaiming B0. The two part-2 cases start independently from this same initial state; any discard of L2 occurs before GC. Count only host-data and GC-copy physical page programs, excluding metadata writes and erases.

1. Show out-of-place update and post-GC mappings.
2. Compute WA without discard and with L2 discarded first.
3. What do discard and crash safety separately require?

OS10-E06 LOOK, SCAN, and wrap cost

Cylinders 0…199, head 53, all requests 98,183,37,122,14,124,65,67 arrive 0. Count physical wrap until the last request completes.

1. Give LOOK-down and SCAN-down paths/totals.
2. Does SSTF total 236 prove it is the same as SCAN?
3. Give upward C-SCAN/C-LOOK totals and starvation premises.

OS10-E07 Parity, write holes, and independent equations

Original RAID 5 stripe: D0=0x3C, D1=0xA5, D2=0x5A, with P=D0⊕D1⊕D2. Parts 1–2 change D1 to 0xA4. Part 3 independently uses the original pre-update stripe and GF(2^8) with modulus 0x11D and Q=D0⊕(2・D1)⊕(4・D2); values and coefficients are hexadecimal.

1. Find old/new parity and reconstruct missing D1.
2. New D1 is durable but parity C3 remains, then D2 fails. Show the wrong reconstruction.
3. Recover D1/D2 with the specified field example.
4. Prove two independent equations suffice. Does duplicating XOR suffice?

OS10-E08 Two-copy crash prefixes

A serialized 512-byte record has a four-byte generation, 504-byte payload and four-byte CRC. C1 and C2 initially hold valid v1. Write and flush C1, then write and flush C2, then ACK. The copies have independent failure domains: writing or tearing one does not damage the other. A successful flush makes the specified version durable under the device contract. Assume this CRC detects each injected torn write, generations do not wrap, and at least one complete copy is not entirely destroyed.

1. Which versions survive a crash during C1 or C2?
2. Give the concrete format and CRC/version selection rule.
3. Prove the invariant and specify repair order under another crash.
4. May unacknowledged work recover new? Is a CPU barrier persistence?

OS10-E09 Topology, interfaces, and swap references

Four disks A/B/C/D. 0+1 stripes AB/CD then mirrors; 1+0 mirrors AB/CD then stripes. A SAN LUN and the following swap example are separate. Slot S7 contains 41 and has exactly two swap-PTE references, from P1 and P2, with no other references or pins. Reuse also requires completion of any in-flight I/O and release of cache ownership.

1. AfterA thenC orB fail, which arrays remain readable?
2. Where doNAS/SAN resolve file offsets? Can two uncoordinated ordinary filesystems mount oneLUN for writes?
3. P1 reads S7 and privately writes 42 while P2 still references S7. May S7 be overwritten/reclaimed?
4. With 512-byte sectors, giveA=[1,101) MiB andB=[101,301) MiB boundaries.

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz, Galvin and Gagne, Operating System Concepts, 9th ed., Chapter 10.

Original educational synthesis by Lin Cing, licensed under CC BY 4.0.

https://github.com/xinhi6716/computer-science-self-study